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Excavator boom extension

Abstract

A pin-on boom extension system for an excavator including a top bar, a bottom boom tube adapted to receive a free end of the excavator boom, the bottom tube including a first aperture aligned with a second aperture, a first securing pin adapted to extend through the first aperture and the second aperture, a first pivot bar secured to the bottom boom tube and to the top bar, the second pivot bar having the first end secured to the bottom boom tube and a second end secured to the top bar, a first adjustment plate extending from the first side of the bottom boom tube to the second side of the bottom boom tube, and a second adjustment plate extending from the first side of the bottom boom tube to the second side of the bottom boom tube.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1898453	12/1932	Lehman	212/175	E02F 3/382
4641439	12/1986	Shirek	414/739	E02F 3/382
5993139	12/1998	Deneve	414/722	E02F 3/382
6116847	12/1999	Ginn	414/722	E02F 3/283
9970177	12/2017	Koestler	N/A	E02F 3/365
10066363	12/2017	McGee, Jr.	N/A	F16C 33/20
11035094	12/2020	Ferguson	N/A	E02F 3/3405
2011/0150618	12/2010	Ramun	414/722	E02F 3/965
2016/0021823	12/2015	Hoppel	280/98	E02F 3/3677

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3553231	12/2018	EP	A01D 87/0076
2224483	12/1989	GB	E02F 3/3405

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/413,853 filed on Oct. 17, 2022 and titled EXCAVATOR BOOM EXTENSION. The content of this application is incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present invention relates to systems and methods for providing a boom extension for heavy machinery. More specifically, the present invention provides a pin-on boom extension that can be added to a mini-excavator between the existing boom and the attachment.
BACKGROUND OF THE INVENTION
(2) A mini-excavator is a construction machine that can perform a variety of tasks such as digging and loading. A mini-excavator may be used by an operator positioned in the cab of the mini-excavator. The operator may drive the mini-excavator to a desired position and then operate the hydraulic boom and attachment of the mini-excavator to perform the desired function. The mini-excavator may need to be repositioned if the boom is not of a sufficient length to reach areas in which work needs to be done. Repositioning the mini-excavator can be time consuming and may

damage the surface on which the mini-excavator is operating. This damage may be of particular concern when the mini-excavator is operated on sod. Therefore, a need exists for a device to extend the reach of a mini-excavator without the need to reposition the mini-excavator.
(3) This background information is provided to reveal information believed by the applicant to be of possible relevance to the present invention. No admission is necessarily intended, nor should be construed, that any of the preceding information constitutes prior art against the present invention.
SUMMARY OF THE INVENTION
(4) With the above in mind, embodiments of the present invention are related to a pin-on boom extension system for an excavator. The excavator may have a boom and an attachment. The pin-on boom extension system may include a top bar, bottom boom tube, first securing pin, first pivot bar, second pivot bar, first adjustment plate, and second adjustment plate.
(5) The top bar may have an attachment end and an opposing boom end.
(6) The bottom boom tube may have an attachment end and an opposing boom end, which may be adapted to receive a free end of the boom. The bottom boom tube may include a first aperture and a second aperture. The first aperture may have a center and be located on a first side of the bottom boom tube between a first end of a first pivot bar and the attachment end of the bottom boom tube. The second aperture may have a center and be located on a second side of the bottom boom tube between a first end of a second pivot bar and the attachment end of the bottom boom tube. The center of the first aperture may be aligned with the center of the second aperture.
(7) The first securing pin may be adapted to extend through the first aperture of the bottom boom tube and the second aperture of the bottom boom tube.
(8) The first pivot bar may have the first end secured to the first side of the boom end of the bottom boom tube and a second end secured to the boom end of the top bar.
(9) The second pivot bar may have the first end secured to the second side of the boom end of the bottom boom tube and a second end secured to the boom end of the top bar.
(10) The first adjustment plate may have a lower surface and an opposing upper surface. The first adjustment plate may be positioned at an upper portion of the bottom boom tube proximate the boom end and may extend from the first side of the bottom boom tube to the second side of the bottom boom tube.
(11) The second adjustment plate may have a lower surface and an opposing upper surface positioned proximate the lower surface of the first adjustment plate. The second adjustment plate may be positioned at a lower portion of the bottom boom tube proximate the boom end and may extend from the first side of the bottom boom tube to the second side of the bottom boom tube.
(12) The bottom boom tube may include a third aperture, a fourth aperture, and a second securing pin. The third aperture may have a center and be located on a first side of the bottom boom tube between the first aperture and the attachment end of the bottom boom tube. The fourth aperture may have a center located on a second side of the bottom boom tube between the second aperture and the attachment end of the bottom boom tube. The second securing pin may be adapted to extend through the third aperture of the bottom boom tube and the fourth aperture of the bottom boom tube. The center of the third aperture may be aligned with the center of the fourth aperture.
(13) The bottom boom tube may include a first tube located on the attachment end and extending from the first side of the bottom boom tube to the second side of the bottom boom tube.
(14) The bottom boom tube may include a plate extending downwardly from a lower portion of the bottom boom tube proximate the boom end of the bottom boom tube. The plate may have an aperture extending entirely through a thickness of the plate.
(15) The first pivot bar may include a first aperture and a second aperture. The first aperture may be formed through an entirety of a thickness of the first pivot bar on the first end of the pivot bar. The second aperture may be formed through an entirety of the thickness of the first pivot bar on the second end of the pivot bar.
(16) The first adjustment plate may include a first threaded aperture and a second threaded

aperture. The first threaded aperture may be located through an entirety of a thickness of the first adjustment plate. The second threaded aperture may be located through an entirety of the thickness of the first adjustment plate.

(17) The first adjustment plate may include a third aperture located through an entirety of the thickness of the first adjustment plate.

(18) The first adjustment plate may include a fourth threaded aperture and a fifth threaded aperture. The fourth threaded aperture may be located through an entirety of a thickness of the first adjustment plate. The fifth threaded aperture may be located through an entirety of the thickness of the first adjustment plate.

(19) The second adjustment plate may include a first threaded aperture and a second threaded aperture. The first threaded aperture may be located through an entirety of a thickness of the second adjustment plate. The second threaded aperture may be located through an entirety of the thickness of the second adjustment plate.

(20) The boom extension system may include a first wear plate, which may be positioned between the first adjustment plate and the second adjustment plate.

(21) The boom extension system may include a second wear plate, which may be positioned between the second adjustment plate and the first wear plate.

(22) The first wear plate may include a pin extending orthogonally from a surface of the first wear plate proximate the first adjustment plate. The pin may extend through an alignment hole disposed through an entirety of a thickness of the first adjustment plate.

(23) The pivot bar may be rotatably secured to the boom end of the bottom bar and the boom end of the top bar.

(24) The top bar may be rotatably secured to first pivot bar and rotatably secured to the second pivot bar.

(25) The top bar may include a first tube and a second tube. The first tube may have a center located at the boom end. The second tube may have a center located at the boom end. The center of the first tube may be aligned with the center of the second tube.

(26) The top bar may include an elongated cylindrical member and a spreader bar. The spreader bar may have a first end and an opposing second end and a first side and an opposing second side. The first side of the spreader bar may be secured to a boom end of the elongated cylindrical member across a transverse axis of the elongated cylindrical member. An outer surface of the first tube may be secured to the second side of the spreader bar at the first end of the spreader bar. An outer surface of the second tube may be secured to the second side of the spreader bar at the second end of the spreader bar.

(27) A notch may be formed along the second side of the spreader bar between the first tube and the second tube.

(28) The top bar may have an aperture located at the attachment end, extending through an entirety of a thickness of the top bar from a first side to an opposing second side.

(29) The top bar may include a threaded aperture and a threaded rod. The threaded aperture may be located at the attachment end of the top bar along a longitudinal axis of the top bar. The threaded rod may have a first end carried by the threaded aperture, a second, opposing end, and a securing member located on the second end of the threaded rod.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Some embodiments of the present invention are illustrated as an example and are not limited by the figures of the accompanying drawings, in which like references may indicate similar elements.

(2) FIG. **1** is a perspective view of a pin-on boom extension system according to an embodiment of

the present invention.
(3) FIG. **2** is a top plan view of the pin-on boom extension system of FIG. **1**.
(4) FIG. **3** is a side elevation view of the pin-on boom extension system of FIG. **1**.
(5) FIG. **4** is a rear elevation view of the pin-on boom extension system of FIG. **1**.
(6) FIG. **5** is a front elevation view of the pin-on boom extension system of FIG. **1**.
(7) FIG. **6** is a side elevation view of a top bar of the pin-on boom extension system as depicted in FIG. **1**.
(8) FIG. **7** is a top plan view of the top bar of FIG. **6**.
(9) FIG. **8** is a side elevation view of a top bar of the pin-on boom extension system according to an embodiment of the invention.
(10) FIG. **9** is a top plan view of a wear plate of the pin-on boom extension system as depicted in FIG. **1**.
(11) FIG. **10***a* is a top plan view of an adjustment plate of the pin-on boom extension system as depicted in FIG. **1**.
(12) FIG. **10***b* is a top plan view of an adjustment plate of the pin-on boom extension system in accordance with another embodiment of the present invention.
(13) FIG. **11** is a side elevation view of the plate assembly of the pin-on boom extension as depicted in FIG. **1**.
(14) FIG. **12** is a perspective view of the prior art excavator.

DETAILED DESCRIPTION OF THE INVENTION

(15) The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. Those of ordinary skill in the art realize that the following descriptions of the embodiments of the present invention are illustrative and are not intended to be limiting in any way. Other embodiments of the present invention will readily suggest themselves to such skilled persons having the benefit of this disclosure. Like numbers refer to like elements throughout.
(16) Although the following detailed description contains many specifics for the purposes of illustration, anyone of ordinary skill in the art will appreciate that many variations and alterations to the following details are within the scope of the invention. Accordingly, the following embodiments of the invention are set forth without any loss of generality to, and without imposing limitations upon, the claimed invention.
(17) In this detailed description of the present invention, a person skilled in the art should note that directional terms, such as “above,” “below,” “upper,” “lower,” and other like terms are used for the convenience of the reader in reference to the drawings. Also, a person skilled in the art should notice this description may contain other terminology to convey position, orientation, and direction without departing from the principles of the present invention.
(18) Furthermore, in this detailed description, a person skilled in the art should note that quantitative qualifying terms such as “generally,” “substantially,” “mostly,” and other terms are used, in general, to mean that the referred to object, characteristic, or quality constitutes a majority of the subject of the reference. The meaning of any of these terms is dependent upon the context within which it is used, and the meaning may be expressly modified.
(19) An embodiment of the invention, as shown and described by the various figures and accompanying text, provides a pin-on boom extension system **100** for an excavator **200**, specifically, a mini-excavator. As depicted in FIG. **12**, known excavators may have booms **210** of fixed length. An attachment **220** may be secured to the boom **210**. One attachment **220** may be removed and replaced by a different attachment **220** using a known pin-on system. The pin-on boom extension system **100** may be secured to the excavator **200** between the existing boom **210**

and the attachment **220**. An attachment end of the pin-on boom extension system **100** may be secured to the attachment **220**, which may be, by way of example, and not as a limitation, a bucket, auger drive, brush cutter, mulcher concrete breaker, trencher, pallet fork, plate compactor, ripper, packer wheel, or the like. The pin-on boom extension system **100** may be quickly and easily removed from or attached to the boom **210**.

(20) The pin-on boom extension system **100** may include a top bar **110**, bottom boom tube **120**, pivot bars **130**, **180**, and adjustment plates **140**.

(21) The top bar **110** may have an attachment end **111** and a boom end **112**. The top bar **110** may be an elongate member with the attachment end **111** opposing the boom end **112** and a length of the top bar **110** disposed between the two. An attachment end **111** of the top bar **110** may be configured to secure to an attachment **220**. A boom end **112** of the top bar **110** may be secured to a second end **132** of a first pivot bar **130** and a second end **182** of a second pivot bar **180**.

(22) There may be at least two pivot bars **130**, **180**. The pivot bars may each have a first end **131**, **181** and an opposing second end **132**, **182**. A first end **131** of the first pivot bar **130** may be secured to the bottom boom tube **120**. Specifically, the first end **131** of the first pivot bar **130** may be secured to a first side **127** of the boom end **122** of the bottom boom tube **120**. A first end **181** of the second pivot bar **180** may also be secured to the bottom boom tube **120**. Specifically, the first end **181** of the second pivot bar **180** may be secured to the second side **128** of the boom end **122** of the bottom boom tube **120**. A second end **132** of the first pivot bar **130** may be secured to the boom end **112** of the top bar **110**. A second end **182** of the second pivot bar **180** may be secured to the boom end **112** of the top bar **110**. Both the bottom boom tube **120** and the top bar **110** may be rotatably secured to the pivot bars **130**, **180**. Such a configuration may allow the attachment end **111** of the top bar **110** and the attachment end **121** of the bottom boom tube **120** to pivot relative to the pivot bars **130**, **180**.

(23) A first tube **194** may be secured to the boom end **112** of the top bar **110**. The length of the first tube **194** may run perpendicular to the length of the top bar **110**. A second tube **195** may be secured to the boom end **112** of the top bar **110**. The length of the second tube **195** may run perpendicular to the length of the top bar **110**. A center of the void defined by the first tube **194** may align with a center of the void defined by the second tube **195**. A fastener may be carried by the first tube **194** to secure the top bar **110** to the first pivot bar **130**. A different fastener may be carried by the second tube **195** to secure the top bar **110** to the second pivot bar **180**.

(24) The top bar **110** may include an elongated cylindrical member **113**, which may extend from the attachment end **111** to the boom end **112** of the top bar **110**. A spreader bar **114** may be secured to a boom end **119** of the elongated cylindrical member **113**. The spreader bar **113** may be a planar structure and may be secured to the elongated cylindrical member **113** parallel to a plane intersecting a longitudinal axis of the elongated cylindrical member **113**. The spreader bar **113** may be centered on a cross-section of the elongated cylindrical member **113**. The spreader bar **113** may have a first end **115** and an opposing second end **116**. The first end **115** and second end **116** may be separated by a length of the spreader bar **113**. The spreader bar **116** may also have a first side **117** and an opposing second side **118**. The first side **117** of the spread bar may extend from the first end **115** to the second end **116**. The second side **118** of the spreader bar **114** may also extend from the first end **115** to the second end **116**. The first side **117** and second side **118** may have equal lengths. The lengths of the first side **117** and second side **118** may be longer than the lengths of the first end **115** and second end **116**. The first side **117** of the spreader bar may be secured to the elongated cylindrical member **113** or carried by the elongated cylindrical member **113** at the boom end **119**. The first side **117** of the spreader bar **114** may be positioned closer to the attachment end **111** of the top bar **110** than the second side **118** of the spreader bar **114**. An outer surface of the first tube **194** may be secured to the second side **118** of the spreader bar **114** at the first end **115** of the spreader bar **114**. An outer surface of the second tube **195** may be secured to the second side **118** of the spreader bar **114** at the second end **116** of the spreader bar **114**. A notch **196** may be formed along

the second side **118** of the spreader bar **114** between the first end **115** and the second end **116** of the spreader bar **114**. The notch **196** may be positioned between the first tube **194** and the second tube **195**.

(25) The top bar **110** may have an aperture **197** located in an attachment end **111** of the top bar **110**. The aperture **197** may extend through an entirety of a thickness of the top bar **110** from a first side to an opposing second side. The aperture may be formed through a plane bisecting a longitudinal axis of the top bar **110**. The aperture **197** may be configured to receive a pin to secure to the attachment **220**.

(26) In one embodiment, the effective length of the top bar **110** may be adjustable. In such an embodiment, as depicted at least in FIG. **8**, a threaded aperture **198** may be formed in the attachment end of the elongated cylindrical member **113**. The threaded aperture **198** may extend along a longitudinal axis of the elongated cylindrical member **113**. The threaded aperture **198** may not extend through an entirety of the length of the elongated cylindrical member **113** but may have an opening at the attachment end **111** of the top bar **110**. The threaded aperture **198** may be adapted to receive a threaded rod **199**. The threaded rod **199** may have a first end that is carried by the threaded aperture **198**. The length of the threaded rod **199** protruding from the threaded aperture **198** may be adjusted by positioning the threaded rod **199** either further within or further without the threaded aperture **198** as desired. A second end of the threaded rod **199**, which opposes the first end, may be positioned outside of the threaded aperture. The second end may include a securing member **183**, which may have an aperture. This securing member **183**, may be utilized to secure an attachment **220** to the top bar **110**.

(27) A first aperture **133** may be formed through an entirety of a thickness of either pivot bar **130**, **180** on the first end **131**, **181** of the pivot bars **130**, **180**. A fastener may be positioned though each of the first apertures **133** to secure the pivot bars **130**, **180** to the bottom boom bar **120**. A second aperture **134** may be formed through an entirety of a thickness of the either pivot bar **130**, **180** on the second end **132**, **182** of the pivot bars **130**, **180**. A fastener may be positioned though each of the second apertures **134** to secure the pivot bars **130**, **180** to the top bar **110**.

(28) The bottom boom tube **120** may have an attachment end **121**, a boom end **122** opposing the attachment end **121**, and an elongate portion extending between the two ends **121**, **122**. The boom end **122** of the bottom boom tube **120** may be configured to receive a free end of a boom **210** of an excavator **200**. The free end of the boom **210** may be received in a void defined by sides of the bottom boom tube **120** at the boom end **122**. The boom end **122** of the bottom boom tube **120** may have a first side **127** opposing a second side **128**. Both the first side **127** and second side **128** may extend from an upper portion **123** of the bottom boom tube **120** to a lower portion **124** of the bottom boom tube **120**.

(29) The first pivot bar **130** may secure to an outside of the first side **127** and the second pivot bar **180** may secure to an outside of the second side **128**. The inside of the first side **127** and the inside of the second side **128** may define the void within which the boom **210** may be carried. Both the first side **127** and second side **128** may be formed from elongate members that extend down a longitudinal axis of the bottom boom tube **120**.

(30) A first aperture **125** may be formed in the first side **127** and a second aperture **126** may be formed in the second side **128**. Both the first aperture **125** and the second aperture **126** may be arcuate and they may have centers that aligned with one another along a plane projected on a longitudinal axis of the bottom boom tube **120**. The first aperture **125** may be located through an entirety of a thickness of the first side **127** at a position between the connection point of the first pivot bar **130** to the bottom boom tube **120** and the attachment end **121** of the bottom boom tube **120**. The second aperture **126** may be located through an entirety of a thickness of the second side **128** at a position between the connection point of the second pivot bar **180** to the bottom boom tube **120** and the attachment end **121** of the bottom boom tube **120**. The first aperture **125** and second aperture **126** may be sized to allow a securing pin **186** to be carried by, inserted into, and removed

from both apertures **125**, **126**. The securing pin **186** may be inserted into the apertures **125**, **126** and pass through an aperture in the boom **210** to secure the pin-on boom extension system **100** to the excavator **200**. The securing pin **186** may have an aperture to receive a cotter pin and secure the securing pin **186** within the first aperture **125** and second aperture **126**.

(31) A third aperture **191** may be formed in the first side **127** of the bottom boom tube **120** and a fourth aperture **192** may be formed in the second side **128** of the bottom boom tube **120**. Both the third aperture **191** and the fourth aperture **192** may be arcuate and they may have centers that aligned with one another along a plane projected on a longitudinal axis of the bottom boom tube **120**. The third aperture **191** may be located through an entirety of a thickness of the first side **127** at a position between the first aperture **125** and the attachment end **121** of the bottom boom tube **120**. The fourth aperture **192** may be located through an entirety of a thickness of the second side **128** at a position between the second aperture **126** and the attachment end **121** of the bottom boom tube **120**. The third aperture **191** and fourth aperture **192** may be sized to allow a securing pin **187** to be carried by, inserted into, and removed from both apertures **191**, **192**. The second securing pin **187** may be inserted into the apertures **191**, **192** and pass through an aperture in the attachment **220** to secure the pin-on boom extension system **100** to the attachment **220**. The second securing pin **187** may have an aperture to receive a cotter pin and secure the securing pin **187** within the third aperture **191** and fourth aperture **192**.

(32) The bottom boom tube **120** may also include a tube **193**, which may be positioned at the attachment end **121** of the bottom boom tube **120**. The tube **193**, may be cylindrical and extend from a first side **127** to a second side **128** of the bottom boom tube **120**. The length of the tube **193** may extend along a transverse axis of the bottom boom tube **129**. The tube **193** may be sized to receive a portion of an attachment **220** and function cooperatively with the third aperture **191** and fourth aperture **192** to secure the attachment **220** to the pin-on boom extension system **100**.

(33) The bottom boom tube **120** may also include a plate **184** that extends down from a lower portion **124** of the bottom boom tube **120**. The plate **184** may be a planar structure positioned in a plane parallel to a plane bisecting a longitudinal axis of the bottom boom tube **120**. The plate **184** may be positioned closer to the boom end **122** of the bottom boom tube **120** than the attachment end **121** of the bottom boom tube **120**. The plate **184** may have an aperture **185** formed through an entirety of a thickness of the plate **183**. The aperture **184** may be positioned between the connection point of the pivot bar **130** to the bottom boom tube **120** and the first aperture **125**. The aperture **185** may be formed so the void of the aperture **125** is parallel to the first side **127** and second side **128** of the bottom boom tube **120**.

(34) An adjustment plate **140** may be secured to the bottom boom tube **120**. The adjustment plate **140** may be a planar member having an upper surface **142** and an opposing lower surface **141**. The adjustment plate **140** may be positioned at an upper portion **123** of the bottom boom tube **120** and extend from the first side **127** of the bottom boom tube **120** to the second side **128** of the bottom boom tube **120**. The adjustment plate **140** may be positioned proximate the boom end **122** of the bottom boom tube. There may be one or more apertures **143**, **144**, **146**, **147** formed through an entirety of a thickness of the adjustment plate **140** from the upper surface **142** to the lower surface **141**. One or more of these apertures **143**, **144**, **146**, **147** may be threaded and adapted to receive a bolt. Some embodiments of the pin-on boom extension system **100** may have two of the apertures **143**, **144**, as depicted in FIG. **10***b*. Other embodiments may have four of the apertures **143**, **144**, **146**, **147**, as depicted in FIG. **10***a*. In embodiments in which one or more of the apertures **143**, **144**, **146**, **147** are threaded, a bolt may be received by on each of the apertures **143**, **144**, **146**, **147** present on the adjustment plate **140**. The bolts may be adjusted independently of one another with a head end of the bolt proximate the upper surface **142** of the adjustment plate **140** and the opposing, terminal end of the bolt extending through the thickness of the adjustment plate **140** to contact with a portion of the boom **210** carried by the bottom boom tube **120**. In embodiments that include a wear plate **150**, **170**, the bolts may contact the wear plates **150**, **170** rather than the boom **210**.

Adjustment of the bolts will place pressure on the boom **210**, either directly or through the wear plates **150**, **170**, and form an interference fit to position the boom **210** within the bottom boom tube **210** at an operable position.
(35) A fifth aperture **145** (or third aperture **145** in embodiment with only two bolt apertures **143**, **144**) may be formed through an entirety of the thickness of the adjustment plate **140**. This aperture **145** may be configured to receive an alignment pin **151** located on a wear plate **150**.
(36) An second adjustment plate **160** may be secured to the bottom boom tube **120** opposing the first adjustment plate **140**. The second adjustment plate **160** may be essentially identical to the first adjustment plate **140**. The second adjustment plate **160** may also be a planar member having an upper surface **162** and an opposing lower surface **161**. The second adjustment plate **160** may be positioned at a lower portion **124** of the bottom boom tube **120** and extend from the first side **127** of the bottom boom tube **120** to the second side **128** of the bottom boom tube **120**. The second adjustment plate **160** may be positioned proximate the boom end **122** of the bottom boom tube. The upper surface **162** of the second adjustment plate **160** may opposed the lower surface **141** of the first adjustment plate **140**. There may be one or more apertures **143**, **144**, **146**, **147** formed through an entirety of a thickness of the second adjustment plate **160** from the upper surface **162** to the lower surface **161**. One or more of these apertures **143**, **144**, **146**, **147** may be threaded and adapted to receive a bolt. Some embodiments of the pin-on boom extension system **100** may have two of the apertures **143**, **144**, as depicted in FIG. **10***b*. Other embodiments may have four of the apertures **143**, **144**, **146**, **147**, as depicted in FIG. **10***a*. In embodiments in which one or more of the apertures **143**, **144**, **146**, **147** are threaded, a bolt may be received by on each of the apertures **143**, **144**, **146**, **147** present on the second adjustment plate **160**. The bolts may be adjusted similarly to the bolts of the first adjustment plate **140**. Adjustment of the bolts will place pressure on the boom **210**, either directly or through a wear plate **150**, **170**, and form an interference fit to position the boom **210** within the bottom boom tube **210** at an operable position.
(37) A fifth aperture **145** (or third aperture **145** in embodiment with only two bolt apertures **143**, **144**) may be formed through an entirety of the thickness of the second adjustment plate **160**. This aperture **145** may be configured to receive an alignment pin **151** located on a second wear plate **170**.
(38) A first wear plate **150** may be positioned between the first adjustment plate **140** and the boom **210** when the boom is carried by the bottom boom tube **120**. The first wear plate **150** may be a planar member. The first wear plate **150** may be proximate the first adjustment plate **140** and located between the first side **127** and second side **128** of the bottom boom tube **120**, between the first adjustment plate **140** and second adjustment plate **160**, and along the lower surface **141** of the first adjustment plate **140**. The upper surface of the first wear plate **150** may contact the lower surface **141** of the first adjustment plate **140**. The first wear plate **150** may include a pin **151** extending upwardly from an upper surface of the first wear plate **150**, orthogonal to the plane of the wear plate **150**. The pin **151** may extend through an aperture **145** formed in the adjustment plate **140**. A cotter pin may be secured through an aperture in the pin **151** to secure the wear plate **150** to the adjustment plate **140**. The wear plate **150** may advantageously receive friction from the bolts passing through the first adjustment plate **140** to put pressure on the boom **210** carried within the bottom boom tube **120**. The friction caused by one or more of these bolts may erode portions of the wear plate **150**. The wear plate **150** may be removed and replaced from the pin-on boom extension system **100** by simply removing and replacing the cotter pin securing the wear plate **150** to the adjustment plate **140**.
(39) A second wear plate **170** may be positioned between the second adjustment plate **160** and the boom **210** when the boom is carried by the bottom boom tube **120**. The second wear plate **170** may also be a planar member. The second wear plate **170** may be proximate the second adjustment plate **160** and located between the first side **127** and second side **128** of the bottom boom tube **120**, between the first adjustment plate **140** and second adjustment plate **160**, and along the upper

surface **161** of the second adjustment plate **160**. The lower surface of the second wear plate **170** may contact the upper surface **162** of the second adjustment plate **160**. The second wear plate **170** may include a pin extending downwardly from a lower surface of the second wear plate **170** orthogonal to the plane of the wear plate **170**. The pin may extend through an aperture formed in the adjustment plate. A cotter pin may be secured through an aperture in the pin to secure the second wear plate **170** to the second adjustment plate **160**. The second wear plate **170** may advantageously receive friction from the bolts passing through the second adjustment plate **160** to put pressure on the boom **210** carried within the bottom boom tube **120**. The friction caused by one or more of these bolts may erode portions of the wear plate **170**. The wear plate **170** may be removed and replaced from the pin-on boom extension system **100** by simply removing and replacing the cotter pin securing the wear plate **170** to the adjustment plate **160**.
(40) Some of the illustrative aspects of the present invention may be advantageous in solving the problems herein described and other problems not discussed which are discoverable by a skilled artisan.
(41) While the above description contains much specificity, these should not be construed as limitations on the scope of any embodiment, but as exemplifications of the presented embodiments thereof. Many other ramifications and variations are possible within the teachings of the various embodiments. While the invention has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best or only mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited. Moreover, the use of the terms first, second, etc. do not denote any order or importance, but rather the terms first, second, etc. are used to distinguish one element from another. Furthermore, the use of the terms a, an, etc. do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item.
(42) Thus the scope of the invention should be determined by the appended claims and their legal equivalents, and not by the examples given.

Claims

1. A pin-on boom extension system for an excavator having a boom and an attachment, wherein the boom extension system comprises: a top bar having an attachment end and an opposing boom end; a bottom boom tube having an attachment end and an opposing boom end adapted to receive a free end of the boom, further comprising: a first aperture having a center and located on a first side of the bottom boom tube between a first end of a first pivot bar and the attachment end of the bottom boom tube, and a second aperture having a center located on a second side of the bottom boom tube between a first end of a second pivot bar and the attachment end of the bottom boom tube, and wherein the center of the first aperture is aligned with the center of the second aperture; a first securing pin adapted to extend through the first aperture of the bottom boom tube and the second aperture of the bottom boom tube; the first pivot bar having the first end secured to the first side of the boom end of the bottom boom tube and a second end secured to the boom end of the top bar; the second pivot bar having the first end secured to the second side of the boom end of the bottom boom tube and a second end secured to the boom end of the top bar; a first adjustment plate having

a lower surface and an opposing upper surface, wherein the first adjustment plate is positioned at an upper portion of the bottom boom tube proximate the boom end and extending from the first side of the bottom boom tube to the second side of the bottom boom tube; and a second adjustment plate having a lower surface and an opposing upper surface positioned proximate the lower surface of the first adjustment plate, wherein the second adjustment plate is positioned at a lower portion of the bottom boom tube proximate the boom end and extending from the first side of the bottom boom tube to the second side of the bottom boom tube.

2. The boom extension system of claim 1 wherein the bottom boom tube further comprises: a third aperture having a center and located on a first side of the bottom boom tube between the first aperture and the attachment end of the bottom boom tube; a fourth aperture having a center located on a second side of the bottom boom tube between the second aperture and the attachment end of the bottom boom tube; and a second securing pin adapted to extend through the third aperture of the bottom boom tube and the fourth aperture of the bottom boom tube; and wherein the center of the third aperture is aligned with the center of the fourth aperture.

3. The boom extension system of claim 1 wherein the bottom boom tube further comprises a first tube located on the attachment end and extending from the first side of the bottom boom tube to the second side of the bottom boom tube.

4. The boom extension system of claim 1 wherein the bottom boom tube further comprises a plate extending downwardly from a lower portion of the bottom boom tube proximate the boom end of the bottom boom tube, wherein the plate has an aperture extending entirely through a thickness of the plate.

5. The boom extension system of claim 1 wherein the first pivot bar comprises: a first aperture formed through an entirety of a thickness of the first pivot bar on the first end of the pivot bar; and a second aperture formed through an entirety of the thickness of the first pivot bar on the second end of the pivot bar.

6. The boom extension system of claim 1 wherein the first adjustment plate further comprises: a first threaded aperture located through an entirety of a thickness of the first adjustment plate; and a second threaded aperture located through an entirety of the thickness of the first adjustment plate.

7. The boom extension system of claim 6 wherein the first adjustment plate further comprises: a third aperture located through an entirety of the thickness of the first adjustment plate.

8. The boom extension system of claim 7 wherein the first adjustment plate further comprises: a fourth threaded aperture located through an entirety of a thickness of the first adjustment plate; and a fifth threaded aperture located through an entirety of the thickness of the first adjustment plate.

9. The boom extension system of claim 6 wherein the second adjustment plate further comprises: a first threaded aperture located through an entirety of a thickness of the second adjustment plate; and a second threaded aperture located through an entirety of the thickness of the second adjustment plate.

10. The boom extension system of claim 1 further comprising: a first wear plate positioned between the first adjustment plate and the second adjustment plate.

11. The boom extension system of claim 10 further comprising: a second wear plate positioned between the second adjustment plate and the first wear plate.

12. The boom extension system of claim 10 wherein the first wear plate includes a pin extending orthogonally from a surface of the first wear plate proximate the first adjustment plate and wherein the pin extends through an alignment hole disposed through an entirety of a thickness of the first adjustment plate.

13. The boom extension system of claim 1 wherein the pivot bar is rotatably secured to the boom end of the bottom bar and the boom end of the top bar.

14. The boom extension system of claim 1 wherein the top bar is rotatably secured to first pivot bar and rotatably secured to the second pivot bar.

15. The boom extension system of claim 14 wherein the top bar comprises: a first tube having a

center located at the boom end; and a second tube having a center located at the boom end; and wherein the center of the first tube is aligned with the center of the second tube.

16. The boom extension system of claim 15 wherein the top bar further comprises: an elongated cylindrical member; and a spreader bar having a first end and an opposing second end and a first side and an opposing second side wherein the first side of the spreader bar is secured to a boom end of the elongated cylindrical member across a transverse axis of the elongated cylindrical member; and wherein an outer surface of the first tube is secured to the second side of the spreader bar at the first end of the spreader bar and an outer surface of the second tube is secured to the second side of the spreader bar at the second end of the spreader bar.

17. The boom extension system of claim 16 wherein a notch is formed along the second side of the spreader bar between the first tube and the second tube.

18. The boom extension system of claim 14 wherein the top bar has an aperture located at the attachment end extending through an entirety of a thickness of the top bar from a first side to an opposing second side.

19. The boom extension system of claim 1 wherein the top bar comprises: a threaded aperture located at the attachment end of the top bar along a longitudinal axis of the top bar; and a threaded rod having a first end carried by the threaded aperture, a second, opposing end, and a securing member located on the second end of the threaded rod.
